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MECHATRONIC TECHNICIAN LEVEL 6

ENG/OS/MC/CR/07/6

Manage Mechatronic Projects



TAITA TAVETA NATIONAL POLYTECHNIC

RESEARCH PROPOSAL

PROPOSED TITLE OF THE PROJECT : Stationary Sanitary Pad Segregator

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1 INTRODUCTION

In many institutional settings such as schools, hospitals, and offices, disposal of used sanitary pads remains a sensitive, unhygienic, and costly challenge. External contractors, whose delays often result in overflowing waste, foul odors, and increased risk of infection, empty most disposal bins. Inadequate disposal practices also violate the privacy and dignity of users. This project proposes a self-contained system that automates the shredding and combustion of sanitary pads inside institutional female toilets. The system will be Arduino-controlled and utilize locally sourced materials to reduce costs while improving sanitation.

1.1 BACKGROUND AND CONTEXT

Currently, institutions depend on waste management companies to dispose of used sanitary pads. This process can be slow, inefficient, and unhygienic. The accumulation of waste in public female toilets not only causes discomfort but also poses health and environmental risks. There is a need for an in-situ disposal system that can automatically shred and incinerate pads immediately after use, eliminating dependency on external collection.

1.2 PROBLEM STATEMENT/RESEARCH PROBLEM

This project will offer a hygienic, automated solution to an often-neglected public sanitation problem. By integrating shredding, detection, and GSM communication in one system, institutions can ensure quick and safe disposal of menstrual waste. It will benefit school management by lowering operational costs and benefit users by preserving their dignity and promoting hygiene.

1.3 RESEARCH RATIONALE

To develop an Arduino-based system for automatically shredding and incinerating used sanitary pads in institutional female toilets.

1.4 RESEARCH AIMS, RESEARCH OBJECTIVES AND RESEARCH QUESTIONS

Research Objectives:

- To fabricate a sanitary pad shredder using local materials.
- To implement a bin monitoring system using ultrasonic and current sensors.
- To enable GSM-based alerts when the bin requires combustion.
- To automate the ignition of the incineration chamber using nichrome wire and manual switch control.
- To evaluate the system's effectiveness in improving hygiene and reducing manual waste collection.

Research Questions:

- How can sanitary pad disposal be improved using automation?
- What is the best way to monitor and control the disposal process using sensors and GSM?

- What safety mechanisms are required to ensure safe operation of shredding and combustion?
- How effective is this system in a real-world institutional setup?

2 LITERATURE REVIEW

2.1 KEY CONCEPTS, THEORIES AND STUDIES

Studies in waste management automation reveal the growing use of embedded systems like Arduino to control mechatronic devices in environmental sanitation. Use of ultrasonic sensors for object detection, current sensors for motor activity monitoring, and GSM modules for SMS communication have proven reliable. Previous research in medical waste handling emphasizes safety and privacy, supporting the need for on-site solutions.

2.2 KEY DEBATES AND CONTROVERSIES

There are ongoing debates about the cost-effectiveness, environmental impact, and safety of incineration systems, especially in closed environments. Some researchers support incineration for its speed and efficiency, while others prefer biodegradable solutions. This project favors incineration due to its simplicity, affordability, and ease of ash disposal.

2.3 RESEARCH GAP

While there are projects on general waste shredders and smart dustbins, little attention has been given to the automation of sanitary pad disposal in female toilets. This project seeks to bridge that gap by integrating pad-specific shredding and safe on-site incineration.

3 METHODOLOGY

3.1 RESEARCH DESIGN

The research uses a **quantitative experimental design**, building a working prototype that will be tested in a real-life setting. The system will combine mechanical shredding and electrical ignition through an Arduino-based controller.

3.2 DATA COLLECTION

Data will be collected during prototype testing using:

- Pad count detection via ultrasonic sensor
 - Bin level monitoring via ultrasonic sensor
 - SMS alerts sent to simulate a real user notification
- Observations and performance logs will also be used to track system response and reliability.

3.3 SAMPLING

The sample size includes **one prototype installed in a test washroom**. Observations will be drawn from multiple simulated uses with sanitary pads. Data will be manually recorded during each session for analysis.

3.4 DATA ANALYSIS

The system's reliability and responsiveness will be evaluated through:

- Count of successful shredding operations
 - Accuracy of ultrasonic detection
 - Timeliness of GSM messages
 - Efficiency of combustion (time to burn)
- Simple statistical analysis like success/failure rates will be used.

3.5 PRACTICAL CONSIDERATIONS

- **Power safety:** Stepdown regulators will prevent overvoltage.
- **Burning control:** Nichrome ignition will be manually initiated for safety.
- **Component durability:** Frequent testing will monitor wear and tear.
- **Ethical issues:** No real personal waste will be used—only simulations.

4 PRACTICALITIES

4.1 IMPLICATIONS AND CONTRIBUTIONS TO KNOWLEDGE

4.1.1 PRACTICAL IMPLICATIONS

The project will help institutions reduce operational costs, enhance hygiene in female toilets, and remove reliance on third-party waste disposal.

4.1.2 THEORETICAL IMPLICATIONS

It demonstrates the integration of mechanical, electrical, and communication systems in real-world public health applications using mechatronic principles.

4.2 PROJECT PLAN AND TIMELINES

Week 1: Component sourcing and shredder fabrication

Week 2: Circuit assembly and testing

Week 3: Software programming and GSM integration

Week 4: Testing, documentation, and demonstration Duration.

4.3 BUDGET

Item	Estimated Cost (KES)
Arduino	2000
Power Supply	1500
Dc Motors	2000
L298 H-bridge	500
LCD	800
GSM Module (SIM800L)	1500
LM2596 Step-downs	500
ACS712 Sensor	600
Ultrasonic Sensor	400
Emergency Switch	400
Double Relay Module	500
Nichrome Wire	300
Misc. (wood, metal, wiring)	3000
Total	14000

5 REFERENCES

6 APPENDICES